

Olivier Gilbert

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Education

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|--|----------------------|
| PhD University of Michigan , Astronomy
• Advisor : Dr. Jinyi Yang
• GPA: 4.00/4.00 | Aug 2025 - Present |
| BSc Université Laval , Physics
• CGPA : 4.12/4.33
• Concentration in astrophysics (required 4 specialized elective courses) | Sept 2022 – May 2025 |
| DCS CEGEP de Levis , Natural Sciences
• R-Score : 32.901 | Aug 2020 – May 2022 |

Experience

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|---|---------------------------------------|
| Bishop's University , Research Assistant
Advisor: John J. Ruan
• Searched for signs of binary supermassive black holes in optical light curves.
• Extracted light curves from optical surveys like ZuberCal and ASAS-SN.
• Used gaussian processes to model light curves with quasi-periodic oscillations and red noise.
• Developed software to identify the presence of quasi-periodic oscillation in a light curve. | Sherbrooke, CA
May 2025 – Aug 2025 |
| Bishop's University , Research Assistant
Advisor: John J. Ruan
• Researched morphological link between host galaxies of Quasi-Periodic Eruptions and Tidal Disruption Events.
• Extracted and analyzed images from wide-field surveys.
• Calculated morphological parameters of galaxies from lenstronomy model fitting.
• Used model comparison to determine optimal way to model host galaxies.
• Performed statistical analysis to compare distributions of galaxies.
• Wrote a paper as a first-author. (See the "Publications" section below) | Sherbrooke, CA
May 2024 – Aug 2024 |
| Bishop's University , Research Assistant
Advisor: John J. Ruan
• Researched physics of accretion disks of low-brightness active galactic nuclei by studying changing-look quasars and their spectral energy distribution.
• Analyzed images from the <i>Hubble Space Telescope</i> and the <i>Chandra X-ray Observatory</i> .
• Reduced spectral data using IRAF.
• Modeled quasars using Sérsic ellipse plus point source fitting with PSF convolution.
• Produced paper-quality figures and tables.
• Wrote a paper as a first-author. (See the "Publications" section below) | Sherbrooke, CA
May 2023 – Aug 2023 |
| Université Laval Foundation , Programmer
• Developed and programmed an online generosity calculator 🧮 to fit the income | Québec, CA
2022 |

of the donors.

Publications

Using Faded Changing-Look Quasars to Unveil the Spectral Energy Distribution Evolution of Low-Luminosity Active Galactic Nuclei

Published in ApJ, Apr 2026

Gilbert, O., Ruan, J. J., Duffy, L., Eracleous, M., Anderson, S. F., Green, P. J., Haggard, D., Plotkin, R. M., Runnoe, J. C., Sobolewska, M.

10.48550/arXiv.2508.01933 [↗](#)

A Host Galaxy Morphology Link Between Quasi-Periodic Eruptions and Tidal Disruption Events

Published in ApJ, Dec 2025

Gilbert, O., Ruan, J. J., Eracleous, M., Haggard, D., Runnoe, J. C.

10.48550/arXiv.2409.10486 [↗](#)

Scholarships/Grants

Rackham Conference Travel Grant 25-26, Rackham Graduate School, University of Michigan. \$1150 USD

June 2026

Master's/Precandidate Rackham Graduate Student Research Grant 25-26, Rackham Graduate School, University of Michigan. \$1500 USD

May 2026

Undergraduate Student Research Award (USRA), National Sciences and Engineering Research Council of Canada (NSERC). \$6000 CAD

May 2024 - Aug 2024

Undergraduate Student Research Award (USRA), National Sciences and Engineering Research Council of Canada (NSERC). \$6000 CAD

May 2023 - Aug 2023

Presentations

American Astronomical Society (AAS) 245th Annual Meeting iPoster presentation on Quasi-Periodic Eruptions. NASA/ADS abstract [↗](#)

Jan 2025

Observing

Keck 10m, DEIMOS

2 half-nights

Magellan Baade 6.5m, FIRE

2 nights remote

Skills

Python: numpy, matplotlib, astropy, scipy, lenstronomy, pandas, jax, etc.

Other languages: C++, JavaScript, HTML, CSS, C#

Other tools: LaTeX, SAO DS9, IRAF, HEASARC, CIAO

Leadership and Public Outreach

Presentation at Local Astronomy Club Invited presenter for the monthly meeting of a local astronomy club.

Sep 2024

Basketball: Captain of an Intramural Team at Université Laval

Sep 2023 - Present

Volleyball: Member of an Intramural Team at Université Laval

Sep 2022 - Present

Media Coverage

Astrobites Who's Home? A Supermassive Black Hole & its Tiny Friends [↗](#)

Dec 2024